Thor enclosure test: BigBoy and Slim

October 26, 2026

1 Executive summary

K001 / BigBoy (20×16×12", 44 W/°C HX) and **K002 / Slim** (20×16×8", 22 W/°C HX) ran side by side at the Apiary from 15-26 October 2026 UTC. Both are the same class of Thor with the Avermedia D331 carrier board and ran the same cooler software. The workload included dense and MoE LLM soaks, plus a sustained matmul hold that provides a thermal upper bound for the GPU. Outdoor air ranged from about 5 to 36°C.

Slim remains a candidate on this run. Neither node hit the 55°C interior critical threshold or reported throttling. Slim ran **4.37°C warmer** at simultaneous timestamps, commanded the HX **3.40%** of the time versus **0.45%** on BigBoy (9.45 h vs 1.25 h), and drew more whole-node plug power (mean **+11 W**; **+22 to +61 W** on matched workloads). Because its enclosure is smaller, Slim retained heat longer, ran warmer, called for cooling more often, and drew more power at the AC plug.

- Day interior maxima were similar (40.22°C BigBoy, 39.99°C Slim). At night Slim stayed warmer (34.69°C vs 26.84°C).
- After HX-on commands, both returned to ≤35°C for two continuous minutes. Slim's shorter median (6 vs 15 min) is not faster cooling: it often started much cooler, with a tail to 84 min.
- Reported air dew-point margin stayed above 5°C. That does not clear condensation on lids, walls, or the HX.
- This run is not a 40°C outdoor case, a winter case, or a calibrated sustained heat test, and it does not say whether field power budgets can carry Slim's extra plug watts.

Decision requested: is the evidence enough to accept Slim for Thor, knowing it will usually run hotter, cycle the cooler more, and pull more plug power in the field?

2 Test scope

The two test enclosures are:

- **K001 / BigBoy** as 20×16×12" Altelix NEMA 4X plus an oversized air-to-air Stratus HX (methanol heat pipe, 44 W/°C).
- **K002 / Slim** as 20×16×8" Altelix NEMA 4X with a smaller HX (22 W/°C, about half the heat-pipe cooling, 2" thinner HX).



Figure 1: K001 / BigBoy (left) and K002 / Slim (right) at the Apiary. Same site, post mounting, and outdoor exposure. Each box has its own outdoor sensor. The designed difference is the enclosure size and HX cooling capacity; the Thor, cooler software, and workloads are all kept the same.

Recorded controls for the HX on/off algorithm deployed were identical throughout their available samples: interior on/off **40/34°C**, interior critical **55°C**, SoC-on **75°C**, minimum on/off **90/90 seconds**. The SoC trigger means cooling can start below 40°C interior; the interior thermostat alone does not explain all cycles.

Coverage	BigBoy / K001	Slim / K002
First interior sample, UTC	Oct 15 00:00:00	Oct 15 00:00:00
Last interior sample, UTC	Oct 26 15:12:30	Oct 26 15:13:15
Interior observation hours	279.21	278.06
Interior coverage within each observed span	100.00%	99.58%
Local outdoor range, °C	5.43–35.68	5.27–35.22

The analysis covers data collected from Oct 15 through Oct 27 UTC. Comparisons use the available metric samples. There are **278.04 hours** of simultaneous interior/outdoor observations. No 40°C outdoor test, winter exposure, or calibrated sustained 170 W enclosure heat test is established here.

3 What this run tells us

Both systems operated below the configured 55°C interior critical threshold, with no recorded throttle flags during this run. Slim remains a candidate. Its smaller enclosure is expected to retain heat longer, which would explain warmer operation, more HX activity, and higher whole-node plug power. The three original questions have qualified answers:

Question	Finding
How quickly does interior temperature settle?	After observed HX-on edges, both nodes reached ≤35°C for a continuous two minutes within three hours. Median completion time was 15 min for BigBoy and 6 min for Slim, but Slim often started much cooler and had a long tail to 84 min.
Does Slim cool enough compared with BigBoy?	No interior critical-threshold exposure or reported throttling on either node. Slim was 4.37°C warmer on average at simultaneous timestamps and had 3.40% HX duty versus 0.45%.
What are day/night interior bounds?	Day maxima: 40.22°C BigBoy, 39.99°C Slim. Night maxima: 26.84°C BigBoy, 34.69°C Slim.

4 Thermal performance and control demand

KPI	BigBoy / K001	Slim / K002
Interior mean / p95 / max, °C	20.08 / 34.32 / 40.22	24.44 / 36.09 / 39.99
Observed interior time ≥40°C	7.0 min	0.0 min
Observed interior time ≥55°C	0.00 h	0.00 h
SoC p95 / max, °C	55.47 / 72.06	68.84 / 75.88
HX commanded-on duty	0.45%	3.40%
HX commanded-on time	1.25 h	9.45 h
Reported throttle duty	0.00%	0.00%
Controller sensor-stale duty	0.00%	0.00%

Slim sat warmer for most of the run, not only at the peaks. Mean interior was **4.36°C** higher (24.44 vs 20.08), matching the **4.37°C** simultaneous gap. The p95 values are closer (36.09 vs 34.32) and the maxima are essentially the same (∼40°C) because both controllers share the 40°C on-threshold: heavy soaks drive both boxes into that band. Slim held the ceiling by commanding the HX about eight times as often (3.40% vs 0.45%; 9.45 h vs 1.25 h). BigBoy's 7 minutes at or above 40°C, versus none on Slim, is that extra cooler time, not Slim rejecting heat more effectively.

The chip shows the cost more clearly than interior air. Slim's SoC p95 was **13.4°C** hotter (68.84 vs 55.47), and its maximum **75.88°C** crossed the 75°C SoC-on setpoint, so some Slim cycles started because the processor was hot rather than because interior air had reached 40°C. BigBoy never reached that trigger (max 72.06°C). Neither node hit 55°C interior or reported throttling.

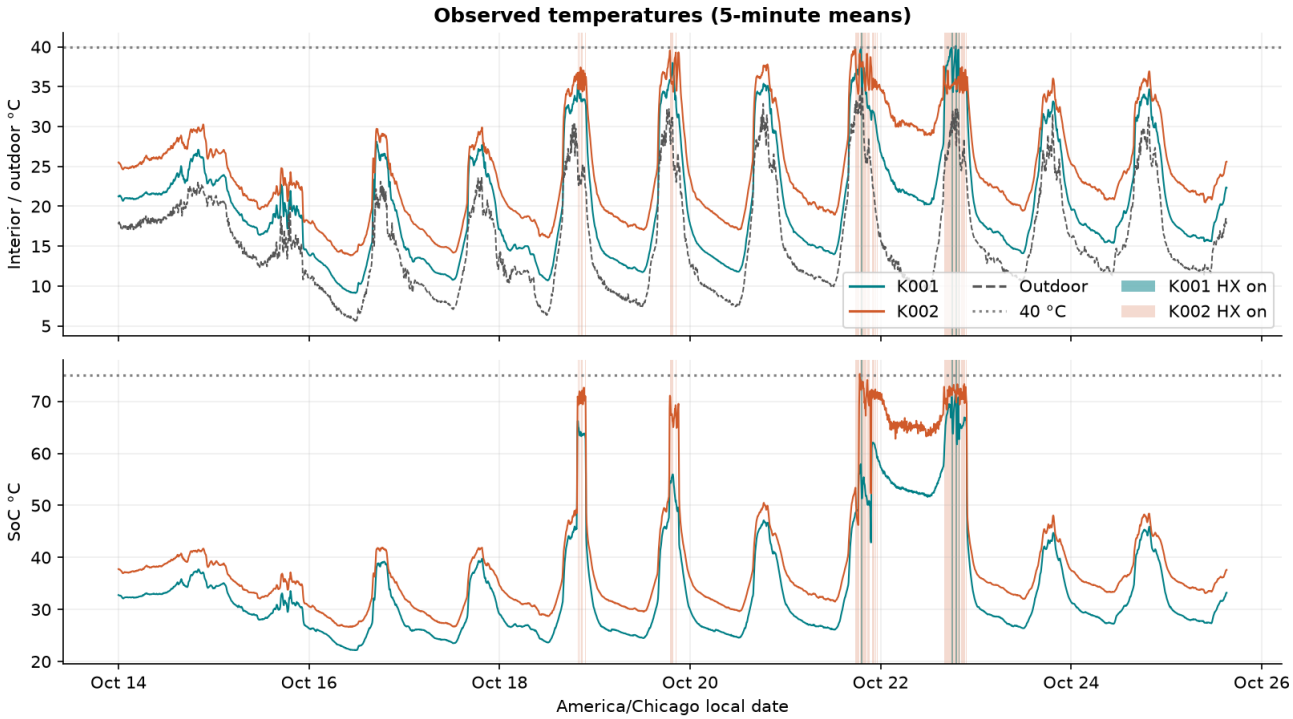


Figure 2: Temperatures and SoC across the run. Shaded bands (node color) mark when that enclosure's HX was commanded on. Five-minute means make the full run readable. The dashed outdoor trace is the mean of the two adjacent outdoor sensors.

Figure 2 shows the gap hour by hour, not only in the summary statistics. Slim (orange) sits above BigBoy (teal) on every day of the run, including 15–18 October when neither HX ran. That offset is the smaller-box floor: both interiors follow outdoor temperature, but Slim does not fall as far overnight. Both approach the 40°C on-threshold during the stress test's afternoons (19, 20, 22, and 23 October). HX-on shading clusters on those peaks and is almost entirely Slim; BigBoy cooling appears only on 22–23 October.

The SoC panel is where the two boxes separate most. During the long 22–23 October GPU hold, Slim's chip stays in a high plateau near 65–75°C and crosses the 75°C SoC-on line, while BigBoy's SoC drops into the 50s between peaks. Interior air looks similar at the ceiling because Slim's cooler is working; processor temperature is what shows the smaller volume costing more.

4.1 Workload comparison

The table reports Slim minus BigBoy at each shared UTC timestamp. Positive values mean Slim was warmer or drew more plug power. The first row uses every paired interior and outdoor sample. Workload rows keep only times when both nodes were running the same labeled job and model. There are no BioCLIP or OpenCV workload labels in this dump.

Workload	Hours	Mean interior difference, °C	Interior difference p05–p95, °C	Mean difference in rise above outdoor, °C	Mean plug power difference, W
All simultaneous	278.04	+4.37	+1.91 to +7.85	+4.47	+11.27
GPU stress / matmul-4096	26.18	+5.45	−3.33 to +8.88	+5.59	+22.06
Dense LLM / gemma3	2.40	+2.49	−2.71 to +7.76	+3.06	+50.42
MoE LLM / gemma4:31b	3.08	+1.46	−2.44 to +6.27	+1.95	+60.65

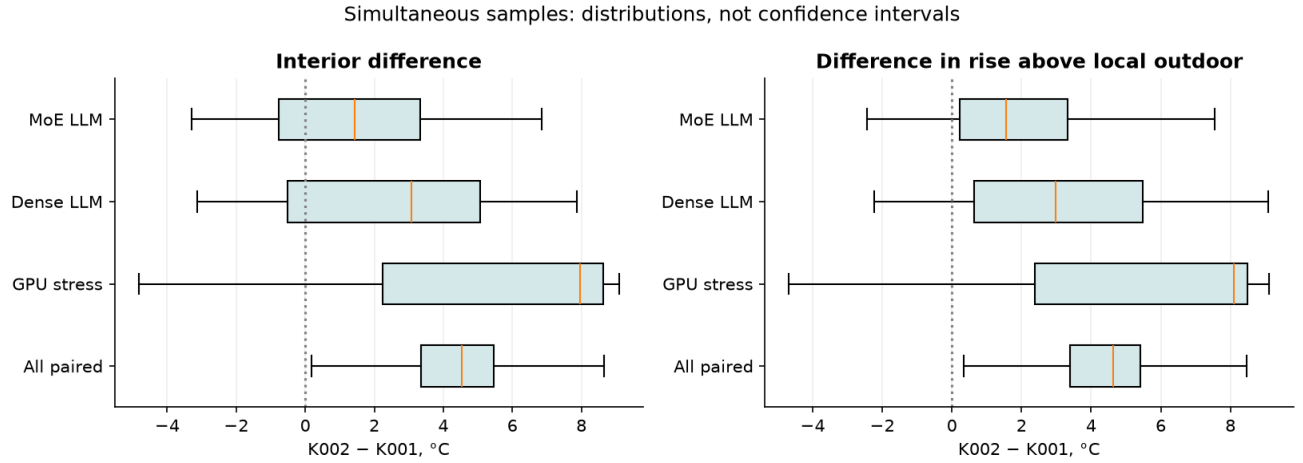


Figure 3: Paired distributions of interior temperature and rise above outdoor. Boxes show the interquartile range and median; whiskers use 1.5×IQR and omit outliers. These distributions are descriptive. Repeated 15-second observations are correlated and are not independent test replications.

The 278-hour all-simultaneous row is the everyday gap: Slim **4.37°C** warmer and **+11 W** on the plug. GPU stress is the only long labeled overlap (26 h) and is warmer still (**+5.45°C**, **+22 W**). Dense and MoE LLM look closer in interior temperature (**+2.49°C** and **+1.46°C**) because Slim's HX is on **41%** and **69%** of those samples versus **0%** and **11%** on BigBoy. The cooler holds Slim's air nearer the shared 40°C band; the cost is plug power (**+50 W** and **+61 W**). Those LLM rows are only 2.4 h and 3.1 h, so they are not a second full test.

Figure 3 is that table as distributions. All-paired and GPU boxes sit to the right of zero; GPU has the highest median and a tail below zero, where Slim is briefly cooler while its HX is running. Dense and MoE boxes straddle zero with lower medians, matching the smaller LLM means. The rise-above-outdoor panel tracks interior temperature because the two outdoor sensors differed by only **0.10°C**: the gap is the enclosure, not the weather.

4.2 Day and night bounds

Local day is 07:00–19:00 America/Chicago; night is 19:00–07:00. The split is clock time, not measured sunlight or a radiative-cooling test. The 22–23 October GPU stress test continued after 19:00, so part of the night sample is still under load.

Table 1: Interior temperature (°C) by node and clock period.

Node / period	Observed hours	Min	p05	Median	p95	Max
BigBoy · day	135.21	9.17	12.18	23.35	35.92	40.22
BigBoy · night	144.00	9.08	11.09	15.72	22.28	26.84
Slim · day	134.46	14.15	17.07	27.71	37.27	39.99
Slim · night	143.60	13.82	14.87	20.90	30.14	34.69

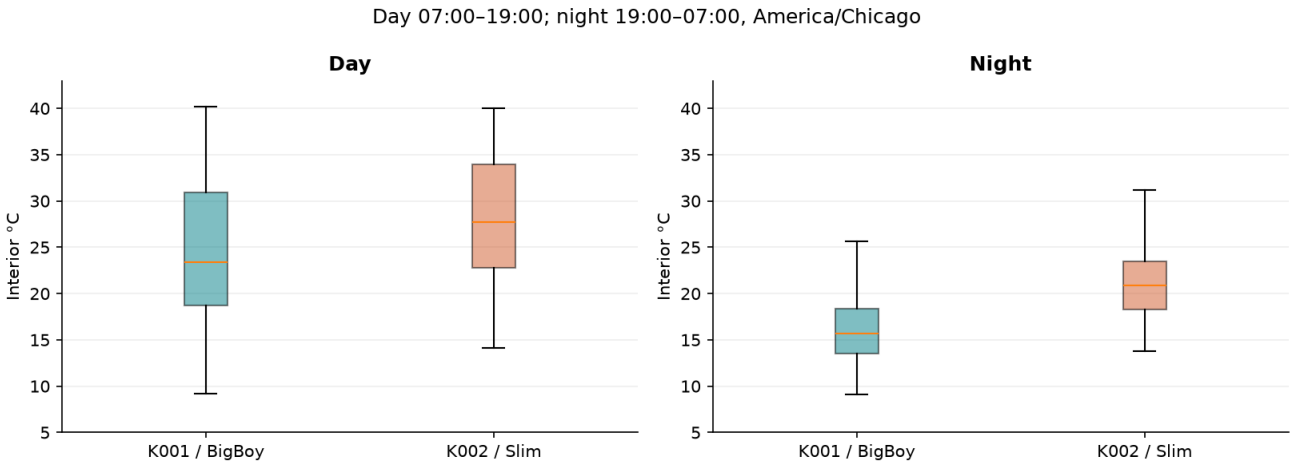


Figure 4: Interior temperature distributions during day and night.

Daytime is when the two boxes look most alike at the top of the range. Medians still differ by **4.36°C** (27.71 vs 23.35), but the maxima are the shared 40°C on-threshold (40.22 vs 39.99) because afternoon soaks and sun drive both interiors into the control band. Figure 4 shows that: both day boxes reach ∼40°C, with Slim's whole distribution shifted up.

Night is where Slim is the warmer enclosure. Its night median is **5.18°C** above BigBoy (20.90 vs 15.72), and p95 and max are about **8°C** higher (30.14 / 34.69 vs 22.28 / 26.84). Slim's night maximum sits at the 34°C off-threshold; BigBoy nights out near 27°C and never needs that band. The night boxes barely overlap. That is the smaller-volume floor from Figure 2: after each day's peak Slim does not dump heat as far, so even its coldest nights (min 13.82°C, p05 14.87°C) stay above BigBoy's (9.08°C, 11.09°C).

5 Recovery after cooling starts

Here, recovery means completing a **continuous 120-second dwell at or below 35°C** ($T_{\text{off}} + 1$°C) after a confirmed off→on command transition. A three-hour search stops at a temperature gap; missing samples never count toward the dwell. The timestamp reported is when the dwell completes. This measures return to a control band, not a fitted thermal time constant or a permanent steady state.

Event measure	BigBoy / K001	Slim / K002
Confirmed off→on starts	4	23
Events reaching the recovery criterion	4 / 4	23 / 23
Median recovery, min	15	6
Recovery range, min	12–18	3–84
Observed on intervals, including gap boundaries	4	24
On intervals censored at a data boundary	0	2

BigBoy's four events began around **39.6–40.1°C**; several Slim events began near **35.7–37°C**. Slim's shorter median therefore cannot be interpreted as faster cooling. The 84-minute recovery during the Oct 22 MoE period shows why the full event distribution matters. Recovery windows can include later cycling or workload changes, so they are not isolated HX experiments. No observed on interval was shorter than 90 seconds, but command cadence cannot validate physical fan operation.

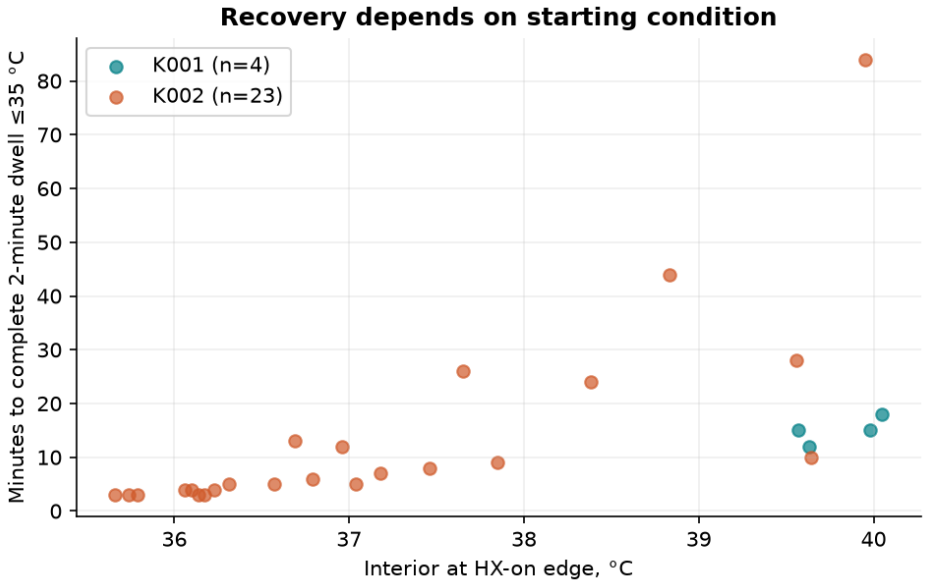


Figure 5: Minutes to finish a continuous two-minute dwell at or below 35°C after each confirmed HX-on command, versus interior temperature at that edge. Each point is one off→on event (4 BigBoy, 23 Slim).

Both nodes have an Oct 19 GPU run, an Oct 20 dense LLM run, and an Oct 22 MoE run followed by a long GPU run. These runs start at different temperatures and change with afternoon weather, so no common workload "settling time" is defensible from them.

6 Workload power

Active label	Node	Plug p50 / p95, W	Board VIN p50, W	Interior max, °C	HX duty
GPU stress	BigBoy	97.30 / 107.04	103.68	40.22	3.49%
GPU stress	Slim	120.49 / 136.76	164.20	38.65	21.61%
Dense LLM	BigBoy	66.81 / 71.54	67.64	38.18	0.00%
Dense LLM	Slim	117.80 / 133.23	120.90	39.98	41.38%
MoE LLM	BigBoy	66.57 / 78.49	65.94	39.99	10.81%
MoE LLM	Slim	136.74 / 140.40	129.09	39.95	69.35%

These summaries use each node's active labeled observations. The paired table above includes only timestamps shared by both nodes. "Unmarked" samples should be assumed idle.

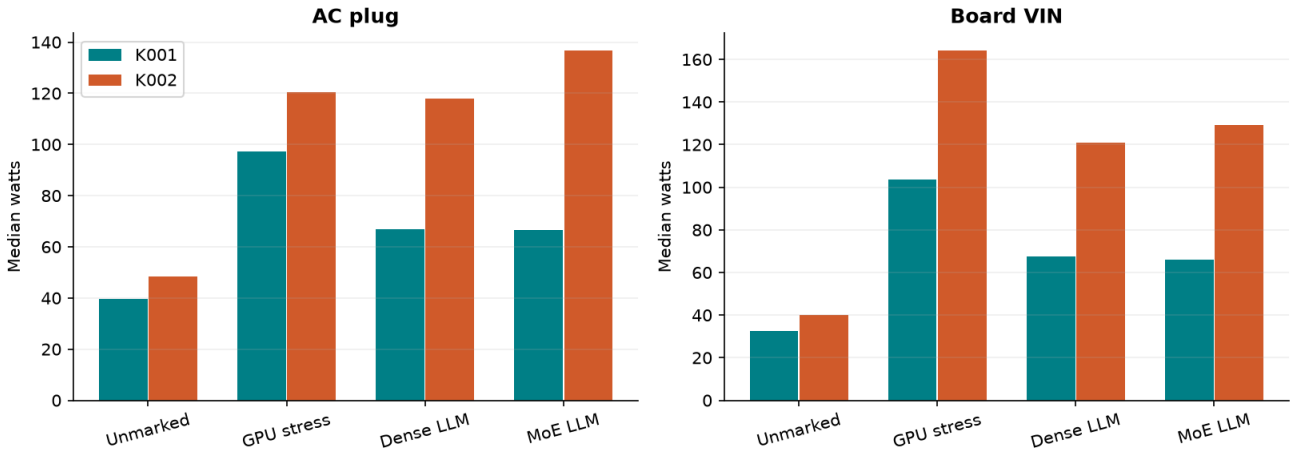


Figure 6: Plug and VIN power by workload.

Slim drew more plug power under the same workload labels. This is consistent with the smaller enclosure driving more HX runtime. The nodes are the same class of hardware and the same cooler software; Slim is a smaller volume, so interior heat takes longer to leave, the box sits hotter, and the HX turns on more. The AC plug is whole-node power (Thor plus other loads on that circuit). It is not board VIN. Extra HX runtime therefore shows up on the plug and will remain true of a smaller enclosure. It is important to note that VIN even exceeds plug in about **8.5%** of overlapping samples on each node; during GPU stress even the median VIN exceeds median plug. Use that disagreement to avoid an instantaneous power/$(T_{\text{in}} - T_{\text{out}})$ rating.

7 Moisture and control reliability

Observation	BigBoy / K001	Slim / K002
Minimum reported air dew-point margin, °C	10.38	8.23
Observed hours with reported margin <5°C / <0°C	0 / 0	0 / 0
Margin observation hours	279.21	278.04
Night observations with interior air below outdoor	0 h	0 h
HX command/PDU overlap	273.18 h	278.06 h
Command/PDU disagreement in that overlap	0 min	1 min

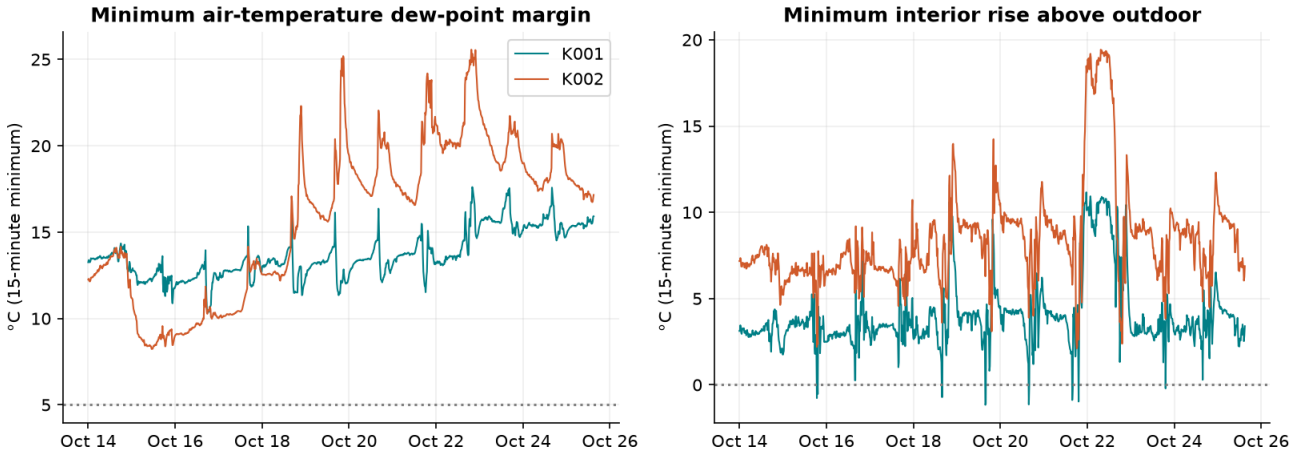


Figure 7: Fifteen-minute minima of air dew-point margin (left; dotted line is the 5°C controller margin) and interior rise above outdoor (right; dotted line is 0°C).

Air dew-point margin stayed well above the 5°C controller floor for the whole dump: minima **10.38°C** on BigBoy and **8.23°C** on Slim, and **zero** hours below 5°C or below 0°C (279 and 278 observation hours). Figure 7 (left) is that floor as a 15-minute minimum. The closest approach is 15–16 October, when Slim dipped toward 8°C on the cool start of the run. After the soaks begin, Slim's margin runs above BigBoy because the smaller box is hotter: $T_{\text{in}} - T_{\text{dew}}$ widens when interior air is warmer. That is the thermal offset from Section 4, not extra moisture protection.

Interior air never went below outdoor (0 h on both nodes). The right panel shows Slim's rise staying several degrees positive and spiking near 19°C during the 22 October GPU hold while BigBoy's 15-minute minimum repeatedly approaches 0°C on cool nights without crossing in the 15-second records.

This does not clear condensation risk. The controller uses the hottest inside air temperature and the wettest inside dew point; a cold lid, wall, HX surface, or connector can reach dew point while that air margin remains positive. Interior air below ambient would be a diagnostic signal, not proof of sky radiation or visible condensation. Add surface temperatures and direct moisture observations for that decision.

8 What are the next steps?

This run does not, by itself, choose an enclosure. Slim stayed under the 55°C interior stop and did not throttle. It ran warmer, used the HX more, and drew more plug power. The questions for the team:

Question	What this run leaves open
Is Slim adequate for Thor, or only adequate in these conditions?	There was no 40°C outdoor case, winter exposure, or calibrated sustained heat load. Day maxima were similar; Slim stayed warmer at night and commanded cooling more often.
Can we support Slim's extra power draw in the field?	Slim used more whole-node plug power (mean +11 W overall; +22 to +61 W on matched workloads) because the HX ran more. That extra load will follow it into deployment. This run does not say whether site power, solar, or battery budgets can carry it.
Can we say which enclosure recovers faster?	Recovery medians are not matched on starting temperature, ambient, or heat input. Slim's shorter median includes starts well below BigBoy's ∼40°C edges.
Have we cleared condensation risk?	Reported air dew-point margin stayed above 5°C, but that is hottest interior air versus wettest interior dew point, not lid, wall, or HX surface temperature.
What would count as a pass?	55°C interior critical and 75°C SoC-on are controller setpoints, not agreed product limits. Headroom, cooling duty, throughput, moisture margin, and footprint are not yet a shared pass/fail rule.

Given that Slim is expected to run warmer, cycle the cooler more often, and draw more plug power, is this evidence enough to accept the Slim enclosure for Thor?

To see the data and tools used to generate this report see here: `https://github.com/waggle-sensor/thor-climate-publisher/tree/stress_testing/benchmarking`